



Heart sound classification method using gammatonegram and SVM

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Abstract

Today cardiovascular diseases have been posing a serious threat to human lives all over the world. Various automated decision-making systems have been proposed by the researchers to help cardiologists to diagnose heart health with more perfection, thus, reducing the frequency of fatalities. This research aims at classifying the Phonocardiogram (PCG) signals captured from the functioning of the cardiac system as normal or abnormal. The methodology proposed here has been validated using the benchmark dataset PhysioNet Computing in Cardiology (CinC) 2016. As of this writing, the studies carried out using contemporary techniques have so far been unable to generate pictorial representation from cardiac audio recordings that match the way the ear processes sound. The present research suggests a unique methodology for the detection of important events in an audio signal using a biologically-inspired depiction of the audio stream through a picture known as Gammatonegram which correlates to the processing of audio in the cochlea membrane of the inner human auditory system. In this study, texture-related features which include Linear Ternary Pattern (LTP), Local Directional Pattern (LDP), Geometric Local Textural Pattern (GLTP), and Local Phase Quantization (LPQ) have been extracted from a visual representation of PCG signal such as Spectrogram, Scalogram, Mel-spectrogram, and Gammatonegram. As compared to the case when researchers were employing other images, it has been noticed that the fusion of the features retrieved from the Gammatonegram has led to an increased overall classification performance metrics. The experiment yielded an overall accuracy of 94.00 % with precision and F1 scores of 91.77 and 93.61 respectively.

Keywords Cardiovascular disease · PCG Signal · Gammatonegram · Support Vector Machine (SVM) · Heart Sound Classification

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1 Introduction

Around 60% of fatalities are believed to be caused by non-communicable diseases such as Cardiovascular Disease (CVD), different cancers, chronic respiratory disorders, and diabetes [1]. CVD is still a major medical concern with a growing monetary cost in low and middle-income countries. About 27.4% of all fatalities are now linked to CVD. In 2019, ischemic heart disease, a particular type of CVD caused 16.17% of total deaths worldwide. It was followed by stroke causing 11.59% of total deaths worldwide [1]. According to the World Burden of Disease Research, there has been a mortality rate of 272 per 100,000 people due to CVD which is more than double the global average of 235.

Connected health technology offers the ability to promote individualized health improvements as well as early illness diagnosis and improved convenience and efficacy in the treatment of chronic diseases [2]. There is much potential to revolutionize CVD prevention and management methods to improve health-related decision outcomes and decrease the expense load of CVD with the increasing availability of a wide variety of smart health devices. In CVD, digital health technology ranges from preventive and disease management to patient engagement and remote monitoring techniques and tools to improve clinicians' abilities for patients [3]. This new area aims at providing safe, effective, and customized attention at all phases of a disease management span. Though today's smart wearable gadgets provide useful help for real-time heart health monitoring, it's difficult to achieve high accuracy in a changing physical environment. The machine learning-assisted diagnosis system incorporated in these smart wearable gadgets interprets the signals gathered by their sensors. One of the challenges in this domain is that signal interpretation algorithms based on machine learning/deep learning have only been verified on a small dataset, therefore, their clinical dependability and diagnostic performance cannot be generalized to real-world clinical applications.

Pulse wave, Electrocardiogram (ECG), Phonocardiogram (PCG), Photoplethysmography (PPG), and Ballistocardiogram (BCG) are the most commonly used non-invasive methods for determining cardiovascular parameters [4]. Despite the fact that auscultation is the standard mechanism for diagnosing cardiac problems, the emphasis has shifted away from it in recent years due to its inaccuracy in determining any abnormality in a heart-beat, as it depends on the hearing skills of a doctor. Therefore, it encouraged a deviation from auscultation to other techniques such as ECG, PCG, and PPG. A comparative study of biomedical signals for heart status detection is presented in Table 1. Since the PCG signals depict heart sounds that are connected with the cardiac mechanical function, they have drawn the interest of clinicians. PCG allows for a more thorough evaluation of features than auscultation does [5, 6].

PCG signal is comprised of four heart audio components (S1, S2, S3, and S4); where, S1 and S2 are interpreted as healthy heartbeats, which are produced when the atrial and ventricle valves of the heart close and open [6]. Besides S1 and S2, a third and fourth sound (S3 and S4) may also be present in the dysfunctional heart. S3 comes right after S2; and S4 appears right before S1 [7]. The four components of the heart sound (S1, S2, S3, and S4), and the heart's diastole and systole are depicted in Fig. 1. During the cycle of one heartbeat, a murmur can be heard making a distinct sound. An abnormal cardiac activity known as a murmur is one of the most prevalent abnormalities. Heart murmurs are distinct from heart sounds as they are louder and last for a longer period of time. Systolic and diastolic heart murmurs can be distinguished by the systolic and diastolic intervals in which they occur.

Table 1 Comparative study of biomedical signals

Signal	Detection Part	Signal Type	Advantage	Disadvantage
ECG	Twelve-lead on chest	Electrical Signal	High accuracy	Contact of electrodes with skin is risky for prolonged application.
PCG	Aortic and Pulmonary valves	Mechanical Signal	Fast, non-invasive, low cost	Consists of noise
PPG	Finger	Photoelectric Signal	Low cost, non-invasive, easy to operate	Susceptible to motion artifacts caused by hand movements
Pulse wave	Wrist	Mechanical Signal	Simple to use	Difficult to accurately monitor
BCG	Sensors integrated within ambient locations such as pillows, chairs, mattresses, bed, etc.	Mechanical Signal	Non-contact	Weak signal

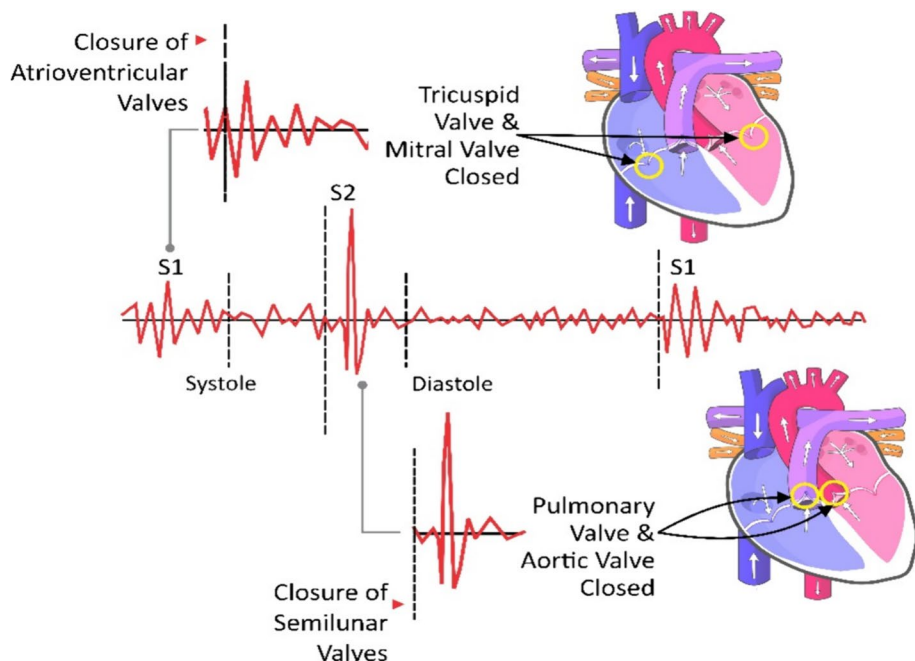


Fig. 1 PCG Signal Representation [8]

1.1 Generic flow of a decision support system

As illustrated in Figure 2, usually, an automated system for classifying heart sounds can be broken down into four distinct levels: data acquisition, pre-processing, feature extraction, and classification into normal and pathological heart sounds.

Several studies have been conducted by including the segmentation phase [9, 10]. This phase is implemented before feature extraction and after the data pre-processing phase, and also by excluding the segmentation phase [11, 12]. Wavelet transform, Fractal decomposition, Hilbert envelope, and Shannon Energy Envelopogram are examples of commonly used cardiovascular-functioning audio recording segmentation techniques. These techniques

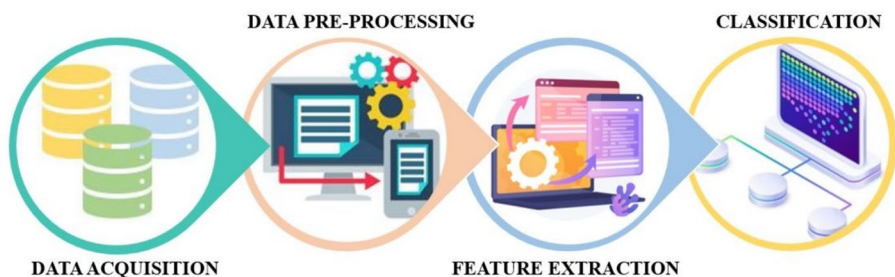


Fig. 2 Stages of Decision-making Support System for Heart Health Diagnosis

function in the time, frequency, or a combination of time-frequency domain, which are also frequently utilised as a foundation for the creation of various segmentation techniques of high complexity [13]. Even though classification performance for segmentation-based PCG features has improved, there are still certain limitations. In some circumstances, ECG recordings—which are difficult to obtain—are also needed for PCG segmentation. There are instances where murmurs cause unusual signal peaks, causing further heart sounds to be overlooked. Patients may also display non-informal murmurs in cases where it is thought that diastole will last longer than systole. Classifying PCG recordings without segmentation allows for a quicker response time and more accurate outcomes [14]. Without performing the segmentation method, Deng et al. [12], extracted features from PCG signals and then used these features to carry out a performance analysis. The results are comparable to those obtained following the segmentation of the heart sound.

Various time [15, 16], frequency [15, 17, 18], and time-frequency features [19, 20] have been derived from PCG signal datasets in many studies (as shown in Fig. 3). Many sound predictive algorithms make use of cepstral features extracted from audio data [21, 22]. These work well in well-ordered environments, but they've failed miserably in noisy ones [12]. In the context of noisy environments, however, characteristics recovered from time-frequency visuals of speech or sound have demonstrated better effectiveness [12], as well as being affected by environmental (music, door closing/opening, *etc.*), and physiological noise (cough, voice, *etc.*). Traditional methods emphasize on frequency while ignoring the need to account for time in some cases which, as a result, makes PCGs prone to signal misinterpretation, resulting in an incorrect diagnosis. Time and frequency isolation are the two advantages of transferring audio data to the visual domain for feature extraction. It is possible that the spectral and temporal information would be maintained by this two-dimensional method. This two-dimensional technique, however, helps to sustain both spectral and temporal intricacies. In the visual domain, the time-frequency representation of

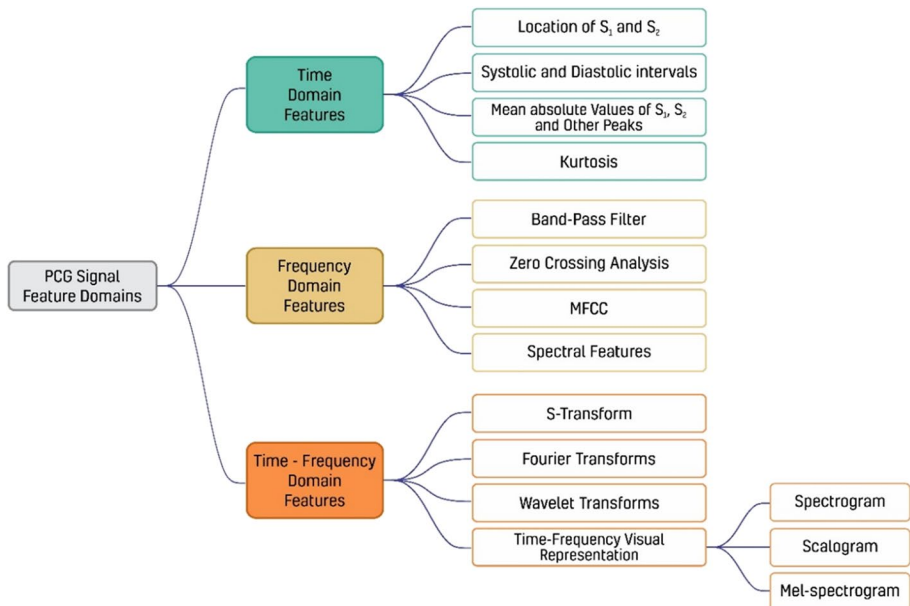


Fig. 3 Features Extracted from PCG Signals through Literature Review

the audio signal frequently presents intriguing patterns. Little attention has been given to audio classification based on visual patterns. Events with short durations but unique time-frequency representations tend to be more common in heart sound recordings [23]. As a result, visual information gleaned from images of the sounds should be useful in classifying them.

1.2 Motivation of this research

Non-stationary heart sound signals necessitate time–frequency domain analysis. Researchers have thus far been unable to produce relevant information from visuals of cardiac sound data that match the way the ear processes sound. When using the response of the gammatone filter, it is appropriate to compare it to other cardiac sound signal visualization techniques because the gammatone filter mimics the human ear's features [24]. The vibration of a structural element in the cochlea of the human auditory system known as the basilar membrane, generates energy that correlates with the frequency of the audio segment that is being heard. The proposed study followed a different yet efficient approach for the identification of important events in an audio signal using a biologically-inspired depiction of the audio stream through an image known as a Gammatonegram, which correlates to the processing of audio in the cochlea membrane of the inner human auditory system [25]. The Gammatonegram shows how an event's frequency distribution changes over time in a certain way. Our approach is to develop a heart health status detection strategy based on image recognition methods because the Gammatonegram image provides the necessary information to detect the audio events of interest. According to the literature review performed by the authors of this work, extraction of Gammatonegram from PCG signals has not been explored.

1.3 Contribution of this research

This research work has contributed significantly in the following manner:

- i. **Audio-Gammatonegram Processing:** This research presents a novel approach that utilizes Gammatonegram to analyse PCG data. The Gammatonegram emulates the auditory processing of the human cochlea, which is a unique approach in the field of heart sound classification. This strategy is novel and significant since it aims to mimic human auditory processing in order to improve the identification of cardiac events.
- ii. **Textural Elements for Feature Extraction:** The study investigates the extraction of textural elements from visual representations of PCG signals, such as Spectrogram, Scalogram, Mel-spectrogram, and Gammatonegram. This approach is significant since it utilizes several forms of data visualizations to enhance the performance of heart sound classification system.
- iii. **Feature Fusion:** A notable aspect is the effective merging of textural features obtained from the Gammatonegram, which has been proven to greatly improve the overall classification performance metrics. This suggests that the combination of features from the proposed representation may be more efficient than utilizing features from other regularly used image representations in similar studies.

2 Literature review

This section offers an overview of the research conducted on classification of heart sounds utilising the information extracted from the heart sound visuals. The methodology involves the conversion of segmented and non-segmented cardiac beat signals into their corresponding visuals before classification. The literature analysis, as presented in Table 2, demonstrates the heart sound classification models where PCG signals have been converted into corresponding images such as Spectrograms, Mel-spectrograms, and Scalograms before their classification.

As neural networks excelled in terms of classification accuracy, an enormous dataset was required to train them. Further, a neural network classifier's method requires an immense amount of time and computational energy to implement, making classification more challenging [55]. However, SVM works well even with a limited dataset for high-dimensional classification problems [56]. Additionally, the performance of the SVM classifier is not inherently correlated with the quantity of specifications present within the input vectors [57]. The SVM classifiers also enable the use of a suitable kernel function, such as a Linear, Gaussian, Polynomial, Radial basis, Morlet wavelet kernel, Mexican hat wavelet, or B-spline, in accordance with the specific cardiac anomaly being studied. It is feasible to optimise the training reliability and model performance by tweaking the parameter values of the kernel function. Due to the given advantages of the SVM classifier, this work has further restricted the literature review to heart sound analysis by translating audio data into images and further classification of heart sound data into abnormal and normal classes using the SVM classifier. Spectrograms, Scalograms, and Mel-Spectrograms are the three forms of visual descriptors/images used in the review of literature.

2.1 Spectrogram

A spectrogram can show the amplitude, frequency, and duration of an audio stream on a single graph. Frequencies keep on changing as can be observed in this graph. In order to indicate amplitude, the third dimension is used, which can be shown either in various shades or color variations. Spectrograms can be generated from audio signals using Fourier Transforms. The use of Fourier Transforms, which breaks down the signal into its constituent frequencies, shows the amplitude of each frequency. For example, a spectrogram breaks an audio file into smaller chunks and utilizes Fourier Transforms to determine which frequencies can be found in each chunk. This figure is created by combining the Fourier Transforms of each of those sections. Colors indicate the amplitude of the frequency in relation to time on the graph. Signals with more energy appear with a stronger color.

In 2024, Zhang et al. [27] presented a heart sound signal classification method combining Mel cepstrum coefficient and second-order spectrum features. These features were then fed into a CNN for classification. Although, the study has demonstrated a significant improvement over traditional methods, leveraging feature fusion for enhanced diagnostic accuracy in heart disease detection, yet the use of complex features such as Mel cepstrum coefficients combined with second-order spectrum and a convolutional neural network requires significant computational resources. This could limit the practical deployment of the model in real-time applications or on portable devices commonly used in clinical or remote environments. Abbas et al. [26] proposed ML and DL techniques to analyse noisy heart sound signals and improve diagnostic accuracy. The study employs data

Table 2 Literature Review of PCG signals converted into corresponding images prior to the binary classification of PCG signals { X_A = PASCAL Classifying Heart Sounds Challenge 2011 dataset A, X_B = PASCAL Classifying Heart Sounds Challenge 2011 dataset B, Y = PhysioNet CinC 2016, Z = INTERSPEECH 2018 ComParE HSS dataset, W = Massachusetts Institute of Technology Heart Sound Database (MITHSDB) and Aalborg University Heart Sound Database (AADHSDB)}

Spectrogram Citation	Year	Features Extracted	Classification Model	Dataset Used	Results	Acc^2	Se^3	Sp^4	UAR^5
Abbas et al. [26]	2024	MFCC features	MLP	$X_A + X_B$	P^1	95.65	97.60	-	-
Zhang et al. [27]	2024	Bi-spectrum heat map and MFCC	CNN	[28]	-	98.50	-	-	-
Singh et al. [29]	2023	Spectrogram	Ensemble of transfer learning models (SqueezeNet + AlexNet + VGG19)	Y	-	99.20	99.47	99.09	-
Tan et al. [30]	2022	Sound Spectrum, MFCC	CNN	[31]	-	81.22	-	-	-
Khan et al. [32]	2021	Time-frequency features	CNN	$X_A + X_B + Y$	95.30	92.70	94.98	89.95	-
Cheng et al. [33]	2019	CNN feature maps	Laconic Neural Network (LNN)	Y	-	Mean=89.50	91.00	88.00	-
Ferdoushi et al. [34]	2019	Spectral centroid, Roll-off frequency, SBW, zero crossings and MFCC features	SVM	X_A X_B	- -	72.22 72.22	- -	- -	- -
Dong et al. [35]	2019	MFCC, log-mel, eGeMAPS	SVM	Z	-	-	-	-	Test = 49.70
Zhang et al. [14]	2017	Tensor features	SVM	X_A	$P_N = 100.00$ $P_{AB} = 64.00$	-	-	-	-
				X_B	$P_N = 51.00$ $P_{AB} = 95.00$	-	-	-	-
				Y	$P_N = 92.00$ $P_{AB} = 88.00$	-	-	-	-

Table 2 (continued)

Zhang et al. [36]	2017	Regression features	SVM	X_A	$P_N=67.00$ $P_M=91.00$ $P_{A^*}=94.00$ $P_E=44.00$	-	100	64	-
Zhang et al. [37]	2017	CNN feature map	SVM	X_B	$P_N=77.00$ $P_M=66.00$ $P_E=67.00$	-	51	90	-
Faurrahman et al. [38]	2017	Deep Belief Network (DBN) feature extraction method	SVM	X_A X_B W	77.00 71.00 -	-	100.00 51.00 -	67.00 80.00 -	-
Singh-Miller and Singh Miller [39]	2016	Mean and variance of all spectral features	Random Forest Regressor	Y	-	-	81.00	89.00	-
Nilanon et al. [40]	2016	Power spectral density features and energy distribution features	Logistic regression (LR) SVM CNN	Y	-	74.20	72.60	75.80	-
Antink et al. [41]	2016	Nonnegative Matrix Factorization (NMF) used to extract spectral components	Random Forest	Y	-	75.60 81.30 78.00	75.60 73.50 74.00	75.60 89.20 83.00	-
Mel-spectrogram									
Citation	Year	Features Extracted	Classification Model	Dataset Used	Results				
Chen et al. [42]	2023	Mel-spectrum and Log Mel-spectrum coefficients	CNN	Y	P -	Acc 71.74	Se 73.86	Sp 70.69	UAR -

Table 2 (continued)

Singh et al. [43]	2023	Mel-spectrogram	U-Net CNN	Y	86.84	96.08	-	96.20	-
Xiang et al. [44]	2023	Mel spectrogram, Log-power spectrogram, Hilbert Envelope	KNN Xgboost SVM Gaussian Naïve Bayes	Y	-	81.76 89.74 83.14 81.76	- - - -	- - - -	-
Nguyen et al. [45]	2022	Log-Mel Spectrogram features	LSTM and CNN	[28]	98.00 (overall)	99.67	98.00 (overall)	-	-
Humayun et al. [46]	2018	Spectral, Cepstral, LLD features, Prosodic features	SVM	Y+Z	-	64.20	-	-	Development = 57.93, Test = 42.00
Amiriparian et al. [47]	2018	ComParE feature set and BoAW features	SVM	Z	-	-	-	-	Test = 47.90
Bozkurt et al. [48]	2018	Sub-band envelope features	CNN	Y	-	81.50	84.50	78.50	-
Scalogram									
Citation	Year	Features Extracted	Classification Model	Dataset Used	Results	P	Acc	Se	Sp UAR
Devi et al. [49]	2023	Wavelet Scattering Transform, Scalogram	KNN, SVM, AlexNet	Y	-	97.19	93.69	98.32	-
Chen et al. [50]	2020	Modified frequency slice wavelet transform (MFSWT) features	CNN	Y	-	93.00	95.00	93.00	-
Alaskar et al. [51]	2019	Basic image features such as blobs, edges, etc. extracted by CNN (AlexNet)	SVM	Y	-	87.65	83.71	90.00	-

Table 2 (continued)

Singh et al. [52]	2019	CNN feature map	AlexNet	Y	-	90.00	90.00	90.00	-
Zhao et al. [53]	2018	VGG16 ImageNet features	SVM	Y	-	Mean=56.20	24.60	87.80	-
Meinijes et al. [54]	2018	CNN features	SVM	Y	-	85.90	87.67	86.10	-

¹Precision
²Accuracy
³Sensitivity
⁴Specificity
⁵Unweighted Average Recall

augmentation and a feature ensembler to enhance the performance of the model. The performance of multilayer perceptron model was the best among several classifiers, underscoring the potential of integrating advanced audio processing and AI techniques in medical diagnostics.

In 2023, Singh et al. [29] used an ensemble-based transfer learning model for predicting imbalanced heart sound signals using spectrogram images. The model employed spectrograms and Short-Time Fourier Transform (STFT) to extract features from PCG data. By using pre-trained convolutional neural network architectures like AlexNet, SqueezeNet, and VGG19, the model was fine-tuned on these images to enhance the predictive performance.

In 2022, Tan et al. [30] presented an approach to sound classification using sound spectrum features instead of MFCC, aligning with privacy considerations in data collection applications like NoiseCapture. The research used the UrbanSound8K dataset and employed both deep learning and machine learning models to assess the efficacy of sound spectrum features in sound classification tasks. The results demonstrated that sound spectrum features achieved comparable results with CNNs and suggested potential areas for further fine-tuning with non-CNN models.

In 2021, Khan et al. [32] proposed a deep learning heart sound classification model using CNN, and its performance was admirable (Table 2). However, there was an absence of an elaborate explanation of the structural schema of CNN model used for classification.

In 2019, Cheng et al. [33] proposed a Laconic Neural Network (LNN) model for heart sound classification using a feature map, extracted by deploying a CNN model for feature extraction which resulted in mean accuracy of 89.50. Ferdoushi et al. [34] extracted certain time-domain features such as zero crossings as well as frequency-domain features, such as roll-off frequency of each frame, Spectral Band Width (SBW), MFCC, and spectral centroids through the STFT method. The authors calculated the accuracy metric of the proposed method for detecting the presence of heart health abnormality on the test set data using a linear SVM classifier. However, Dong et al. [35] proposed feature-fusion techniques of early fusion and late fusion. The extracted features were conjugated just before their loading into the classification model in the early fusion technique. While, in the latter method, firstly, the machine learning models were trained separately using different feature extraction methods; and then the final prognosis was made using a voting process. In this analysis, out of MFCC, formant-based features (F1-F3), and Log Mel feature, the Log Mel feature set outperformed others with a Uniform Average Recall (UAR) of 49.7%.

Two heart sound classification systems were proposed by Zhang et al. in 2017; in each system, the scaled Spectrograms were run through dimension reduction algorithms, such as Principal Component Analysis (PCA) [14] and Partial Least Squares Regression (PLSR) [36] to eliminate irrelevant information. Finally, for classification purpose, the SVM algorithm was implemented. For the purpose of PCG signal segmentation and Spectrogram extraction, Faturrahman et al. [38] used the Shannon energy envelope (SEE) technique and the STFT algorithm respectively. After using PCA for reducing the number of variables in the dataset, the authors moved on to feature extraction with a Deep Belief Network (DBN) before using the support vector machine (SVM) method for classification. Segmenting the PCG signal was an element of the process cycle used by all the aforementioned heart sound classification algorithms. Here, segmentation is a strenuous issue because of the erratic characteristic of the PCG signals and the influence of noise present in heart-functioning audio, such as heart murmurs and other fabricated noises. The fundamental job of classifying PCG signals can be achieved by eliminating the use of heart sound segmentation. The primary goal is to simply examine the heart beat for the presence of any abnormality in it, rather than to figure out which abnormality it is, which necessitates pointing out

the exact location of the heart murmur in a cycle. Similar work has been done by Zhang et al. [37] who proposed a system for classifying unsegmented heart sounds by first extracting spectrograms from PCGs, then performing spectrogram-scaling using the bilinear interpolation. Finally, CNN model was implemented for feature extraction and SVM for classification.

In 2016, after windowing the 5-second segments of the PCG signals, classifiers were used to classify each segment, as proposed by Nilanon et al. [40]. After predicting the class of each segment of the recording, a voting system was set up to determine the final classification result. In this work, the authors extracted Spectrograms and Mel Frequency Cepstral Coefficient (MFCC) features. However, Miller et al. and Antink et al. used random forest as a classification algorithm in their proposed works [39, 41] by extracting spectral components as features. The classification system created here was entirely based on a specific dataset. Consequently, the categorization results obtained were inadequate, and there was enough scope to improve the results further by including the information provided by the frequency bands.

2.2 Mel-spectrogram

As compared to a standard spectrogram, a Mel-spectrogram has a different way to display Frequency versus Time. Firstly, Mel-scale of frequency is utilized as the y-axis in this graph. Secondly, instead of amplitude scale, decibel scale is used to identify the colours.

In 2023, Singh et al. [43] presented an automated diagnostic model for classifying cardiac abnormalities using deep neural networks. It focused on classifying heart-related sounds into normal or pathological categories using the UNet architecture and Mel-spectrograms. The model achieved greater accuracy and specificity on the PhysioNet 2016 dataset, but lower performance on the noisier PhysioNet 2022 dataset. The study emphasized on the potential of deep learning in enhancing diagnostic accuracy without focusing on data cleaning. Xiang et al. [44] discussed the development and evaluation of algorithms for classifying heart sounds using two-dimensional features. The study highlighted that the conversion of heart sound classification to image classification helped in selecting better features crucial for clinical diagnosis. Four different datasets were integrated and pre-processed; and methods like machine learning, deep learning, and transfer learning were employed. It was found that certain features such as logmel and logpower provided better accuracy and consistency in classification as compared to other features like envelope and waveform. Additionally, the application of transfer learning improved the effectiveness of Xception and MobileNet models slightly under experimental conditions. Chen et al. [42], in their proposed study, focused on the classification of heart sounds using CNNs trained with Mel-spectrum and Log Mel-spectrum features derived from heart sound signals. The performance of these features was superior as compared to the traditional handcrafted features for diagnosing cardiovascular diseases. The study explored the effectiveness of different sound processing techniques and found that Log Mel-spectrum features, in particular, improved the classification performance of CNNs, demonstrating their potential for enhancing diagnostic accuracy in varied clinical settings.

In 2022, Nguyen et al. [45] explored the classification of heart sound signals using deep learning models. Two models, long short-term memory (LSTM) and CNN, were developed to classify heart sounds into five categories based on log Mel-spectrogram features. The study demonstrated greater classification accuracies, with CNN showing slightly better performance than LSTM.

In 2018, Humayun et al. [46], and Amiriparian et al. [47] used a deep auto-encoder-based methodology for feature extraction to learn feature representations from Mel-spectrograms such as ComParE and Bag-of-Audio-Words (BoAW) feature sets and some spectral, cepstral and low-level descriptor (LLD) feature sets, which were further utilised to train and test the SVM classifier to predict the cardiac health condition. However, Bozkurt et al. [48] proposed a model in which the authors deployed period marking of S1 and S2 components on PCG signals using ECG signals parallelly, which was not only computationally expensive and time-consuming, but had also resulted in low performance of the model.

2.3 Scalogram

Scalogram is the visual depiction of a signal's Continuous Wavelet Transform (CWT) absolute value in terms of time and frequency. A scalogram should be used as long as the event is of short duration and high frequency, and vice versa.

In 2023, Devi et al. [49] demonstrated that use of wavelet transform methods for feature extraction and application of deep learning models could significantly enhance the classification of heart sounds. However, deep learning models, particularly those like AlexNet, can act as "black boxes," providing little insight into how decisions are made. This lack of transparency can be a significant drawback in medical applications where understanding the rationale behind a diagnosis is crucial for trust and ethical reasons.

In 2020, Chen et al. proposed a PCG signal classification model in which Modified Frequency Slice Wavelet Transform (MFSWT) features were extracted; and CNN was used for the classification phase which performed admirably, but the authors implemented segmentation of heart sound signals which was a time-consuming and computationally expensive process [50].

In 2019, Alaskar et al. [51] tested two methods to extract features. The first method, CNN-based AlexNet, was implemented to extract features, whereas SVM was applied as a classifier. The second method, AlexNet, was implemented to extract features and also used as a classifier. However, the performance of this method was far below the first one. Similarly, AlexNet was used as a classifier by Singh et al. [52] which performed well, but a small portion of the dataset was used for training the AlexNet architecture to classify which could lead to overfitting.

In 2018, Meintjes et al. [54] examined PCG signals by converting them into CWT-based Scalograms and then using the CNN algorithm for classification. The authors observed that when comparing CNN features model to LBP features model with SVM and k-NN classifiers, the former performed better. Zhao et al. extracted VGG16 ImageNet features and deployed an SVM classifier; however, the application of pre-trained ImageNet CNN in other acoustic classification paradigms might have been successful, but in this work, the classification results obtained were not up to the mark [53].

2.4 Research gap

On the basis of literature review undertaken as above, it can be said that the prevailing models have explored the various features extracted from PCG signals. However, a gap exists in effectively integrating non-traditional or less commonly used audio features in the classification models. The previous studies focused on common features like MFCC or spectral features. There is a potential to explore other biologically-inspired features *i.e.*,

Gammatonegram or more complex combinations of existing features that could enhance classification accuracy.

The texture of an image refers to the organization and distribution of its different intensities and distinctive characteristics. Texture features are utilized in many applications such as remotely sensed images, healthcare-related image processing, and content-based image extraction. They have a crucial impact on image processing techniques. It has been observed that a gap lies in the exploration of texture-based feature extraction approaches from images obtained from PCG signals.

3 Methodology

This section delves into all the major processes of the proposed approach, from data gathering to binary data classification, *i.e.*, normal and pathological heart sound classes. Figure 4 here highlights the time-domain PCG Signal of normal and abnormal heart sounds. Figure 5 shows the steps followed in the proposed approach. The significance of the Gammatonegram and textural features used in this research over peer approaches has also been studied.

3.1 Data acquisition

PCG signal datasets derived from Physionet CinC 2016 have been acquired to examine the healthy and diseased patients in clinical and non-clinical situations for the purpose of this study. A total of 3,126 heart sound recordings, ranging from 5 seconds to 120 seconds in length, were included in the Challenge training collection which was divided into five databases (A-E). The dataset details are given in Table 3. The auscultation locations used to record the heart sounds were strategically placed around the body. Most auscultation

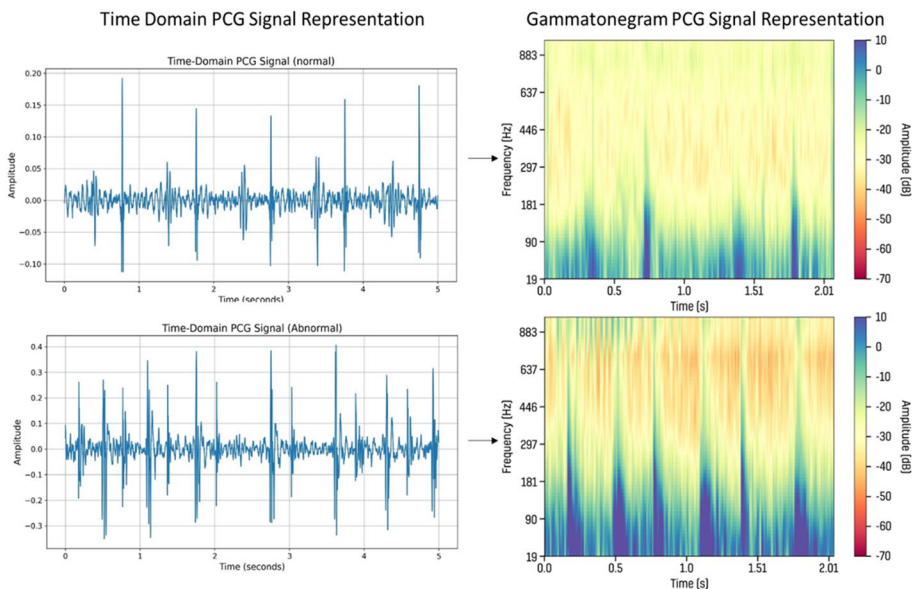


Fig. 4 Time-domain PCG Signal of a Normal Heart Sound and an Abnormal Heart Sound with corresponding Gammatonegram Images

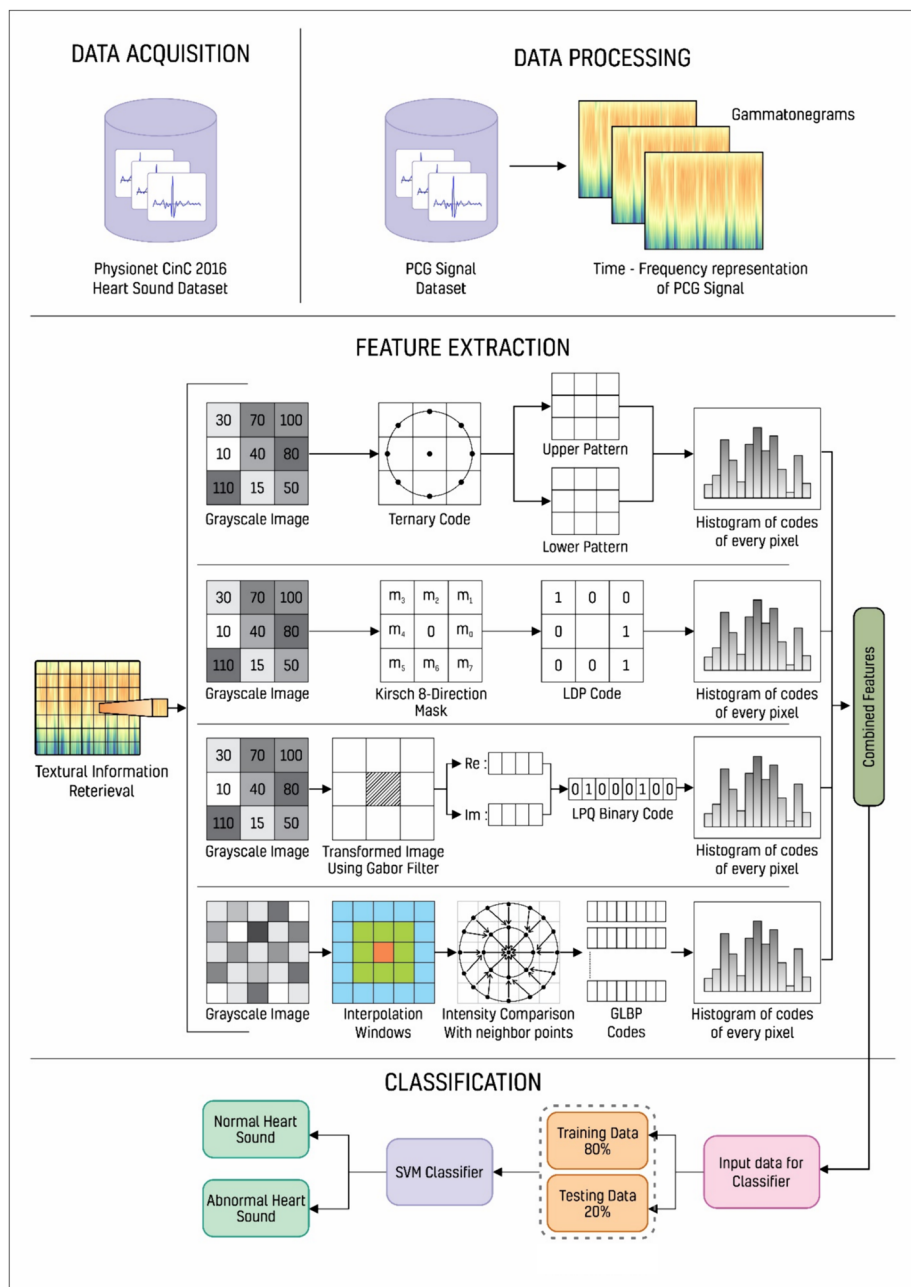


Fig. 5 Methodology Applied in this Work

examinations of heart sounds were focused on the aortic region, pulmonary region, tricuspid region, and mitral region. Both the training and test sets made use of two distinct sets of heart sound recordings; one with a regular heartbeat and the other with an abnormal

Table 3 Physionet/CinC 2016 database details

Types of subjects/classes	Normal and Pathological Signals
No. of recordings	3,126
Age	Variable (children to adult)
Gender	Variable (both male and female)
Sensor positions	9 different locations
Total participants	1072
Duration (seconds)	5–120s each
Sampling rate	2000 Hz
Database access	Open access database

one. The normal recordings related to the persons who were in good health, while people who reported heart issues were the sources of abnormal samples. Thus, both healthy and diseased people were included in the study. Each patient submitted between one and six heart sound recordings. All heart sounds were sampled again to 2,000 Hz and were available in “.wav” files. All the recordings had a single PCG lead.

3.2 Time-frequency visualisation of PCG Signal: gammatonegram

STFT is commonly used to illustrate time-frequency properties. However, the frequency domain of STFT is uniform, whereas the human ear’s reception of varied frequency signals is irregular. Dennis et al. [23] employed a collection of Gammatone filters to generate Gammatonegram to mimic the human ear’s comprehension of sound waves. The Gammatonegram differs from the STFT Spectrogram in that it modulates the bandwidth using a set of Gammatone filters. Each band-pass filter’s center frequency grows correspondingly in the Gammatone filter bank. The resolution of the least frequency difference that a human ear can tolerate is related to the fundamental frequency of a particular sound. Therefore, the perception of frequency difference is irregular. The human ear identifies low-frequency sounds far better than high-frequency sounds. In Gammatonegram, the data dimension that does not contain important sound components can be decreased; and the sound components that are more appealing to the human ear can be highlighted. Fig. 6 demonstrates step-by-step procedure of converting PCG signal audio data into Gammatonegram images. Fig. 7 presents the Gammatonegram of normal and abnormal PCG signals.

The basilar membrane, situated in the cochlea of the human auditory system, generates energy after vibration that correlates to the frequency of the sound that is being heard. The Gammatonegram is an auditory map generated by gammatone filters that mimic the human ear’s natural vibrations. The input heart sound signals are passed through gammatone filters with center frequency f_c to provide energy and are represented in the form of vectors which are then combined to create a Gammatonegram visual representation. These energies are derived by calculating gamma distribution and sinusoidal tone centered at frequency f_c and their product is known as impulse response as shown in Eq. 1; where x , is the amplitude; y is the bandwidth used in gammatone filter; n is the order of filter; and ϕ is the phase shift.

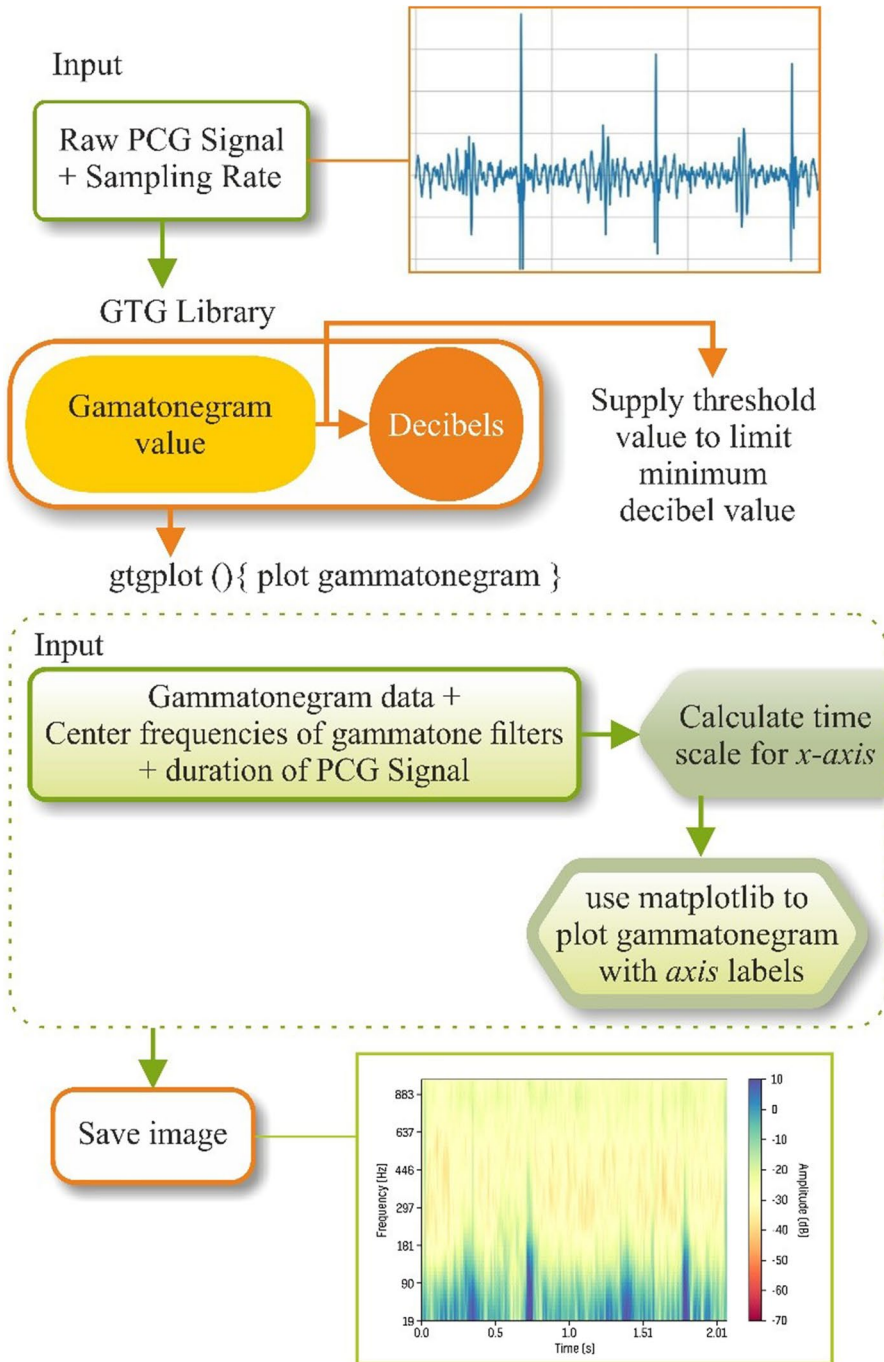


Fig. 6 Block Diagram showing Conversion of PCG Signal to Gammatonegram

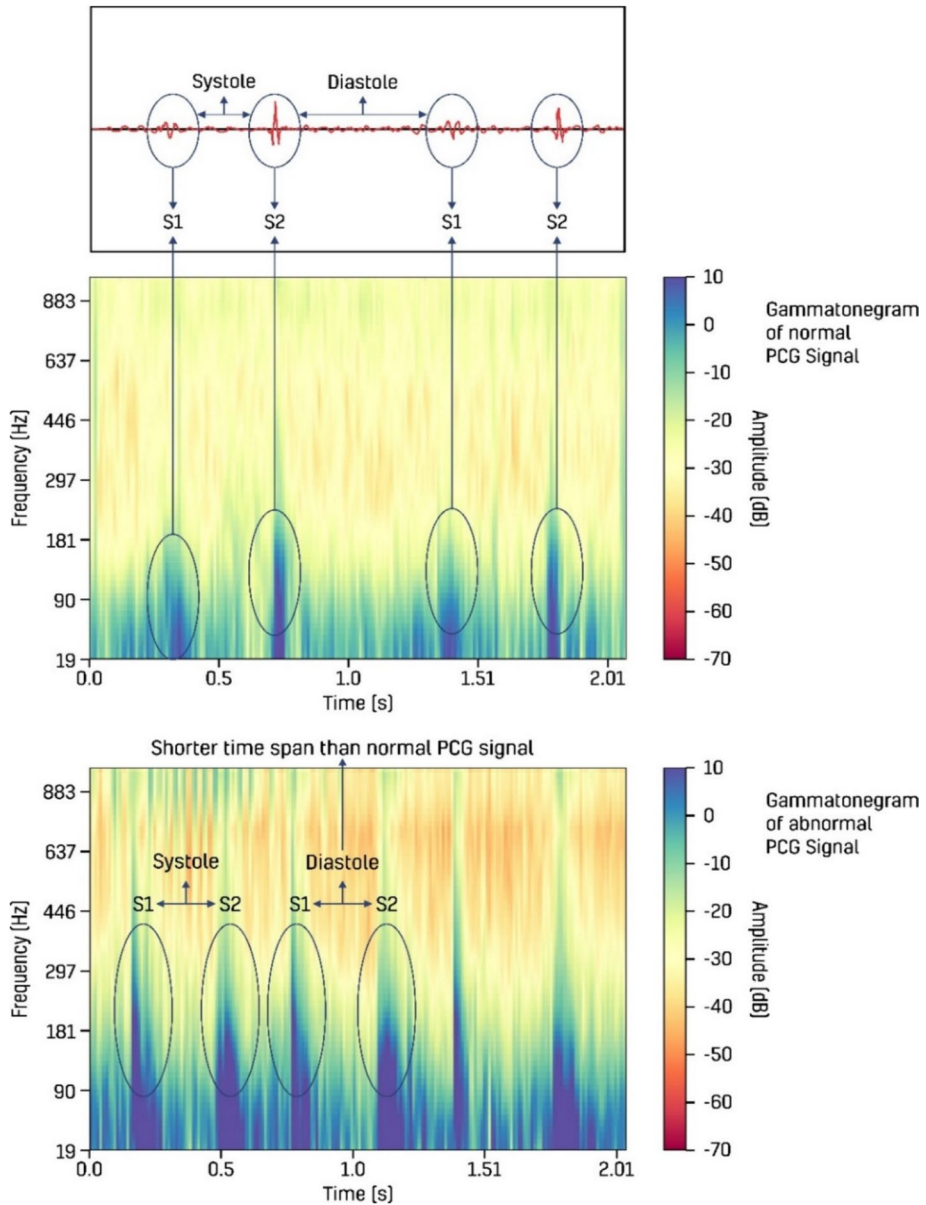


Fig. 7 Gammatonegram of a Normal and Abnormal PCG Signal

$$G_m(t) = xt^{n-1} e^{-2\pi y t} \cos(2\pi f_c t + \phi), \text{ where, } t > 0 \quad (1)$$

In Gammatonegram, the frequencies are dispersed according to the Equivalent Rectangular Bandwidth (ERB) scale as demonstrated in Eq. 2:

$$ERB = 24.7 + 0.108f_c \quad (2)$$

3.3 Extraction of textural features

The texture is a significant aspect of various kinds of images. Texture can be comprehended in images that contain data from remote sensing using many spectral bands as well as minuscule images. Texture information can impact a wide range of machine learning and image recognition applications. As a result, the structural properties of a texture have been the focus of research since the 1960s. Local texture descriptors that are discriminative and computationally efficient, such as Local Binary Patterns (LBPs), have been developed in recent years, resulting in considerable advancements in the implementation of texture approaches to many real-life applications. The LBP texture operator has proven to be a useful technique in a variety of applications, including facial image processing, authentication and identification application such as biometrics, medical pattern classification, motion and progress tracking, and content-based information retrieval from multimedia databases, due to its detection accuracy and computational efficiency. Many other LBP versions have been proposed in order to improve their resilience, discriminative capacity, and application to many sorts of situations. Variants of LBP have also been investigated in this study, which remain to be investigated in the PCG signal classification domain. These have and has demonstrated superior results in comparison to other feature extraction techniques employed by peer approaches for PCG signal classification.

3.3.1 Local Ternary Pattern (LTP)

One of the disadvantages of LBP descriptors is the sensitivity to noise; and heart sound data contain external and physiological noise which causes misreading and poor understanding of the signal, thus, leading to inaccurate diagnosis. The solution to this problem was provided by Tan and Triggs [58] by introducing LTP textural descriptor. The LTP model specifies an image texture by considering three attributes: each pixel c , Q neighbours which are uniformly located at R^1 distance. The LTP operator of a centre pixel (x, y) was derived by discovering a correlation between c and q neighbours as mentioned in Equations 3 and 4, where $q = 1, 2, \dots, Q$ and Q is the total number of neighbours using the neighbourhood radius as R^1 .

$$LTP(x, y) = \sum_{q=0}^{Q-1} s(g_q - g_c) 2^q \quad (3)$$

$$s(g_q - g_c) = \begin{cases} 1, & g_q - g_c \geq \alpha, \\ 0, & -\alpha < g_q - g_c < \alpha \\ -1, & g_q - g_c < -\alpha \end{cases} \quad (4)$$

Where α is a threshold that dictates the change in peak of intensity which must be deemed an edge. Each ternary code, after computation using Eq. 4, is divided into two patterns: a positive (upper pattern) and a negative (lower pattern) code. The pseudo code for extracting LTP features from images is given below in Table 4.

Table 4 Pseudo code for Extracting LTP Features from Images**Pseudo Code: LTP***Input: Gammatonegram image**Output: LTP features**Begin*

1. Import necessary libraries for image processing, numerical computations, data handling, and operating system interactions
2. Define a helper function, *sigmoid*, to classify pixel value differences into three categories:
 - Return '2', if the maximum pixel intensity difference is above a high threshold (150)
 - Return '1', if the maximum pixel intensity difference is below a low threshold (50)
 - Otherwise, return the mean of the pixel intensities
3. Define the main function *LTP* to process an image:
 - a. Initialize an empty matrix (*img_result*) for the LTP image with the same dimensions as the input image
 - b. Loop through each pixel in the input image, avoiding edges where the LTP cannot be applied:
 - i. For each pixel, calculate local ternary values based on the pixel intensity differences with its neighbours:
 - Use the sigmoid function to assign ternary values (0, 1, or 2)
 - Compute these values for eight different neighbour directions around the pixel
 - ii. Combine the ternary values to form the LTP code for the current pixel
 - c. Store the LTP code in the corresponding position of the *img_result* matrix
4. Return the LTP image

*End***3.3.2 Local Directional Pattern (LDP)**

The texture through LDP is defined by calculating the response values at the edges of a pixel in different directions and using these values to cipher the texture of an image. The response values calculated are assigned as a binary code with eight bits of length to each pixel of an input image in the proposed LDP feature. This binary code is ciphered by collating the calculated response values of a pixel in various directions. The edge detection techniques that can be utilized in this regard are the Kirsch edge detection technique, the Prewitt edge detection technique, and the Sobel edge detection technique [59]. The Kirsch edge detector analyses all the eight neighbors, therefore it has been reported to detect distinct edge responses in various directions more reliably than others [60]. As a result, Kirsch edge detector was employed with LDP to extract textural information in this study. The edge response values in eight directions *i.e.* $\{D_i\}$, $i=0,1, \dots, 7$ around a central pixel in the image were calculated using Kirsch masks D_i in eight different ways. In some cases, the response values calculated for each pixel in all the directions were not equally crucial for acquiring necessary information from a texture. Thus, the k most prominent directions were considered to generate the LDP. Here, the most prominent bit responses k_i were set to 1; and the remaining not-so-prominent bits of an 8-bit LDP pattern were set to 0. The pseudo code for extracting LDP features from images is given in Table 5.

$$LDP_K = \sum_{i=0}^7 k_i (D_i - D_K) 2^i \quad (5)$$

Table 5 Pseudo Code for Extracting LDP Features from Images**Pseudo Code: LDP***Input: Gammatonegram image**Output: LDP features**Begin*1. *Import necessary libraries (OS operations, data handling, image processing, numerical operations)*2. *Define the LDP function:*a. *Initialize an empty matrix for the LDP image of the same dimensions as the input image.*b. *For each pixel in the input image:*i. *Define assign_bit function:*- *Determine the binary value based on the comparison of neighbouring pixels in specific directions (0°, 45°, 90°).*- *Assign 0 or 1 based on intensity comparisons.*- *Handle edge cases where neighbouring pixels might be out of bounds.*ii. *Apply the assign_bit function to the current pixel to calculate its new value in the LDP image.*- *Use the sigmoid function to assign ternary values (0, 1, or 2)*- *Compute these values for eight different neighbour directions around the pixel*c. *Return the LDP image.**End*

$$k_i(b) = \begin{cases} 1 & b \geq 0 \\ 0 & b < k \end{cases} \quad (6)$$

3.3.3 Local Phase Quantization (LPQ)

Ojansivu et al. [61] proposed a technique to blur an image which is known as the spatial blurring method. The intensity counts of an image and a point spread function (PSF) are used to depict spatial blurring in the image. The pseudo code for extracting LPQ features from images is given in Table 6. In the context of spatially invariant blurring, the discrete model for the original image $f(x)$ can be described by the convolution as represented in Equations 7 and 8:

$$g(x) = f(x) \otimes h(x) \quad (7)$$

$$G(u) = F(u) \otimes H(u) \quad (8)$$

Where, $h(x)$ is the PSF of the blur image $g(x)$; and $\otimes$ represents a 2-dimensional convolution in Equations 7 and 8. The Discrete Fourier Transforms (DFTs) of $g(x)$, $f(x)$, and $h(x)$ are represented by $G(u)$, $F(u)$, and $H(u)$ respectively. In LPQ, the phase is calculated for N_x neighbourhoods at each pixel positioned at x of the image $f(x)$. The computation of these local spectra using STFT is presented in Eq. 9, where $x \in \{x_1, x_2, \dots, x_N\}$ consists of one-dimensional convolution for the successive rows and columns. The Fourier coefficients are locally calculated at fixed frequency coordinates, $w_1 = [u, 0]^T$, $w_2 = [0, u]^T$, $w_3 = [u, u]^T$, and $w_4 = [u, -u]^T$; where, u is a scalar value to content with condition $H(w_i) > 0$.

Table 6 Pseudo Code for Extracting LPQ Features from Images**Pseudo Code: LPQ***Input: Gammatonegram Image**Output: LPQ features**Begin*

1. Import necessary libraries for numerical computations, data handling, image processing, and plotting
2. Define a function *LocalPhaseQuantization* with parameters for image path, count, and window size:
 - a. Load the image from the specified path
 - b. Convert the image to grayscale to reduce complexity and focus on intensity rather than colour
 - c. Define a parameter 'alpha' derived from the window size, used in the Gaussian derivative for STFT
 - d. Specify convolution mode for edge handling during the filtering process
 - e. Apply the STFT or related filtering technique to extract local features based on phase information
 - f. Plot the processed image based on specified 'count' of features or processing steps
3. Execute the function with appropriate parameters or handle user inputs for interactive processing
4. Return or display the result of the *Local Phase Quantization*

End

$$F(w, x) = \sum_{y \in N_x} f(x - y) e^{-j2\pi w^T y} \quad (9)$$

It is possible to determine the phase information in $F(x)$ by looking at the signs on the real and imaginary sections of each of its components and further computing through simple scalar quantization as given in Eq. 10:

$$q_i = \begin{cases} 1 & \text{if } g_j \geq 0, \\ 0 & \text{otherwise} \end{cases} \quad (10)$$

Where, g_j is the j th component of the vector $G(x) = [Re\{F(x)\}, Im\{F(x)\}]$. Then, the label image $f_{LPQ}(x)$ is defined as:

$$f_{LPQ}(x) = \sum_{j=1}^8 q_j(x) 2^{j-1} \quad (11)$$

3.3.4 Geometric Local Textural Pattern (GLTP)

The basic concept of GLTP feature extraction was initially discussed by Patterson et al. [25] and Tan et al. [58] in their works. The pseudo code for extracting GLTP features from images is given in Table 7. This approach examines the coordinates on the circles with varied radii adjacent to a center pixel are examined to quantify the intensity variation surrounding individual pixels within an image. The intensity value around each pixel is determined through the bilinear interpolation method. Each point on the circular is assigned a bin by comparing its intensity value with other adjacent points on its inner circular region, which is positioned at a smaller radius and has a lower angular difference between the two

Table 7 Pseudo Code for Extracting GLTP Features from Images**Pseudo Code: GLTP***Input: Gammatonegram Image**Output: GLTP features**Begin*

1. Import necessary libraries for image processing, data handling, and operating system interactions
2. Define a function to calculate the gradient of the image using a given template:
 - a. Determine the size of the template (3x3 matrix)
 - b. Create a new image array padded to accommodate the template application without border issues
 - c. Apply the template to compute the gradient across the image:
 - i. Iterate over each pixel in the original image
 - ii. Apply the template to calculate local changes and store the results in the new image
3. Define the main function to process an image for GLTP:
 - a. Initialize an empty matrix for the GLTP image with the same dimensions as the input image
 - b. Loop through each pixel in the input image, avoiding edges where the GLTP cannot be applied:
 - i. For each pixel, calculate gradient values based on the pixel intensity differences with its neighbours using the template:
 - Use the gradient function to assign gradient values
 - Compute these values for multiple directions around the pixel
 - ii. Combine the gradient values to form the GLTP code for the current pixel
 - c. Store the GLTP code in the corresponding position of the GLTP image matrix
4. Return the GLTP image

End

locations. Using the assigned bins of the points on the geometric region around each pixel, one GLTP code is produced. The same GLTP code is allocated to symmetric patterns.

GLTP is built on the idea of oddly numbered neighborhoods that are n-sized and squared-shaped. The center pixel for a neighborhood is calculated as represented in Eq. 12:

$$I_c = \left[\left(n - \frac{1}{2} \right) + 1, \left(n - \frac{1}{2} \right) + 1 \right] \quad (12)$$

The total number of Interpolation Windows (IW) found within the neighborhood is $(n - 1) * (n - 1)$. The computation of a point in an IW generates $(n-1)/2$ neighborhoods in a circular shape around the center pixel, with each computed point belonging to just one circular neighborhood. The positions of the points are determined so that the total distance from the center remains as small as possible. The circular neighborhood radius is computed, as per Eq. 13:

$$\frac{\partial}{\partial x} \sum_{i=1}^n (\sqrt{r^2 - x_i^2} - y_i) = 0 \quad (13)$$

As shown in Eq. 14, P_{ik} is denoted as the intensity value of a point residing in a single IW. P_{ik} is calculated using the bilinear interpolation technique as mentioned in the following equation:

$$P_{ik} = I_{i+1,k} + (I_{i+1,k+1} - I_{i+1,k})x + (I_{i,k} - I_{i+1,k})y + (I_{i,k+1} + I_{i+1,k} - I_{i,k} - I_{i+1,k+1})xy \quad (14)$$

The point within the IW is represented by its Cartesian coordinates x and y . The intensity variations between two points in the GLTP architecture are codified using a Boolean comparison. An intensity value more than or equal to the other point value with which it is being compared will be allocated 1; otherwise, the point is assigned 0.

4 Classification of binary classes

SVM is a robust model used for classification and regression analysis under supervision of the labelled data, which does not rely on probabilistic assumptions, rather it minimises structural risks [62]. It is frequently employed in pattern recognition issues [63, 64]; and in this paper, it has been utilised to categorise the input signal as either "normal" or "abnormal," the two common categories in medical diagnosis. In Eq. 15, $D_{Training}$ is the training data of n size; and X_i is the input data whose labelling is defined by $L_i = 1, -1$ for normal and abnormal classes, respectively.

$$D_{Training} = \{ (X_i, L_i) | X_i \in R^n, L_i \in (\pm 1) \}_{i=1}^n \quad (15)$$

Eq. 16 illustrates how the SVM classifier's n -dimensional hyperplane is represented:

$$f(x) = w \cdot x + b = \sum_{i=1}^n w_i X_i + b \quad (16)$$

Here, w and b are the hyperplane's normal and scalar values respectively; and x is a $n \times 1$ vector. Eq. 17 explains the mathematical expression of a SVM hyperplane:

$$w \cdot X_i + b = \pm 1 \quad (17)$$

The optimal hyperplane sought by the SVM is one that adheres to the performance standards of the model by creating a maximum separation boundary. The linear SVM classifier operates on the input pattern X_i in the training data, where it is described by the parameters w, b . These parameters must adhere to certain rules as mentioned in Eq. 18:

$$\begin{cases} w_i X_i + b > +1, & \text{if } L_i = +1 \\ w_i X_i + b < -1, & \text{if } L_i = -1 \end{cases} \quad (18)$$

The SVM classifier implemented in this work, can be expressed as:

$$f(x) = \sum_{i=1}^n a_i L_i k(x, X_i) + b \quad (19)$$

The given notation represents the class labels as $L_i = (1, -1)$, X_i denotes the input training data, while k represents the kernel function. The other model parameters are denoted by a_i and b . Here, SVM hyperparameter tuning has been done using the Halving Grid Search cross-validation technique. The study utilised the textural feature extraction technique, acquiring LBP-variant features on Gammatonegram images of PCG signals, and subsequently employed an SVM classifier for the purpose of heart sound classification.

5 Experiments and results

A computer system equipped with an Intel Core i9 processor operating at 4.30 GHz, 32 GB of RAM, and a 64-bit operating system was used to conduct all the experiments. Python (version 3.9) was used to implement all the four generic phases of the proposed decision support system, together with the scikit-learn (version 1.1.1) and scikit-image (version 0.18.1) libraries.

The proposed study involves the extraction of Gammatonegram depictions from PCG signals, followed by the extraction of LTP, GLTP, LPQ, and LDP features. A comparative analysis was conducted to assess the impact of information retrieval from Gammatonegram images, in comparison to Spectrogram, Mel-Spectrogram, and Scalogram images. The results obtained from the Gammatonegram image features were superior in comparison to the remaining three images. The SVM algorithm was employed to classify a dataset consisting of two classes, *viz.* abnormal and normal. Halving Grid Search cross-validation technique was applied for hyperparameter tuning selected values of C, gamma, and kernel as 10, 100, and Radial Basis Function (RBF) respectively.

5.1 Performance metrics

The assessment of the classification models was made on the basis of their accuracy, specificity, sensitivity, recall, precision, and F1-score values. The results obtained in this regard are presented in Table 8. In order to derive the performance review scores for the proposed methodology as described in Equations 20–24, the initial step involved the computation of a confusion matrix. A confusion matrix is a metric that demonstrates the efficacy of a decision support system architecture on the testing dataset. A confusion matrix consists of some key terms, *viz.* True Positive (T_P), True Negative (T_N), False Positive (F_P), and False Negative (F_N). Accuracy is a measure that quantifies the proportion of instances correctly categorised to the total number of events.

$$Accuracy = \frac{T_P + T_N}{T_P + T_N + F_P + F_N} \quad (20)$$

$$Sensitivity = \frac{T_P}{T_P + F_N} \quad (21)$$

The sensitivity measure represents the ratio of individuals tested positive through the proposed decision support system to the actual number of patients diagnosed with the disease. In accordance with it, sensitivity was chosen as the primary metric to determine the most efficient methodology when there was a substantial effect on the outcome associated with False Negative. Specificity is the measure of the number of patients predicted negative through the proposed model to the total number of patients having no disease.

$$Specificity = \frac{T_N}{T_N + F_P} \quad (22)$$

Table 8 Results of the proposed research work

Time-Frequency Visual Represen- tation	Textural Features	Testing Accuracy	Specificity	Sensitivity	Precision	F1-Score
Spectrogram	LTP	82.00	64.59	92.53	89.11	90.27
	LDP	83.83	0	100	82.00	89.70
	LPQ	80.43	82.95	80.97	90.21	85.52
	GLTP	82.58	0	100	81.32	89.70
	Fusion	85.87	54.03	94.27	89.65	91.90
Mel-Spectrogram	LTP	90.08	67.18	94.42	92.12	93.25
	LDP	79.20	0	100	80.24	89.04
	LPQ	81.84	82.81	80.38	95.00	87.08
	GLTP	79.88	0	100	80.24	89.04
	Fusion	90.30	75.00	94.20	93.00	92.30
Scalogram	LTP	80.94	68.32	86.43	84.01	85.21
	LDP	79.25	0	100	77.93	87.59
	LPQ	80.84	90.90	80.79	96.91	88.12
	GLTP	80.09	0	100	77.93	87.59
	Fusion	85.283	36.76	93.55	84.77	88.95
Gammatonegram	LTP	83.39	84.00	98.00	84.00	91.00
	LDP	87.25	89.30	97.00	88.00	93.00
	LPQ	89.86	86.40	97.00	91.00	94.00
	GLTP	83.76	79.80	97.00	86.00	91.00
	Fusion (Proposed Method)	93.36	75.60	97.52	91.77	93.61

$$Precision = \frac{T_p}{T_p + F_p} \quad (23)$$

Precision can be defined as the ratio of true positive instances to the sum of true positive and false positive instances. In other words, it quantifies the correctness of a classifier. A high percentage of false positives is indicative of poor precision. The F1-score is a metric used to evaluate the performance of a classifier which considers both precision and recall (also referred to as sensitivity) values to provide a more accurate assessment.

$$F1 \text{ score} = \frac{T_p}{T_p + 1/2(F_p + F_N)} \quad (24)$$

The findings presented in Table 8 reflect that the Gammatonegram produced the most efficient results in comparison to the fusion of textural features across all the images. Consequently, it was decided to employ advanced SVM classifiers (LSSVM, Fuzzy SVM), tree-based classifiers (Decision Tree, Random Forest), KNN, XGBoost, Neural Network models (ANN, CNN), and transfer learning models (ResNet50, DenseNet121, VGG16) for conducting further comparative analysis during the classification phase. The performance metrics of the comparative analysis obtained through the already mentioned algorithms are presented in Table 9. Of all these models, CNN and XGBoost

Table 9 Comparison with other Machine Learning and Deep Learning Algorithms

Algorithms	Testing Accuracy	Specificity	Sensitivity	Precision	F1-score
LSSVM (polynomial)	89.81	74.07	93.20	76.33	75.18
Fuzzy SVM (polynomial)	76.54	76.54	0	0	0
Decision Tree	83.33	57.30	92.25	89.48	90.85
Random Forest	90.58	62.50	96.96	87.00	83.00
XGBoost	91.60	68.79	98.02	91.00	87.00
KNN	88.80	57.80	96.53	80.70	89.32
ANN	90.63	76.92	93.95	93.54	93.97
CNN	91.90	72.46	97.25	90.00	87.00
ResNet50	83.70	68.99	96.09	78.00	67.00
VGG16	80.55	76.71	97.21	76.00	62.00
Densenet121	85.34	68.99	89.40	77.00	78.00

have produced competitive results only, but the performance of the proposed model using SVM classifiers has been the best.

5.2 Receiver operator characteristic curves

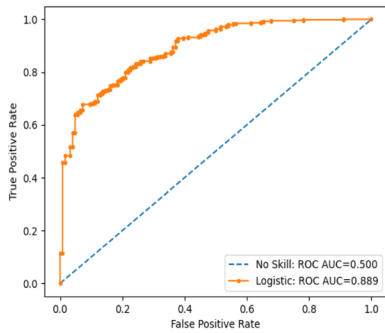
These are commonly employed in binary class labelling scenarios to illustrate the efficacy of a decision support system's algorithm. Fig. 8 displays the ROC curves of the models used for extraction of fusion of textural features, *i.e.*, LTP, LDP, GLTP, and LPQ against Spectrogram, Mel-spectrogram, Scalogram, and Gammatonegram images where the SVM model was utilised for the purpose of classification.

The ROC-AUC score of the proposed model with Gammatonegram image was comparatively higher than Spectrogram, Scalogram, and Mel-spectrogram images. The findings derived from the present investigation have contributed significantly in the area under study.

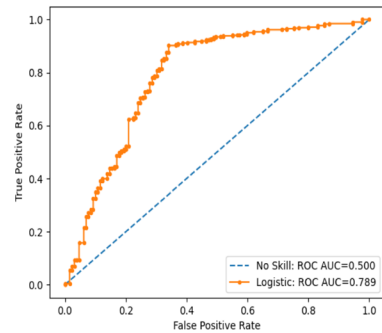
5.3 Result discussion

The results confirming the testing accuracy, specificity, sensitivity, precision, and F1-scores of the proposed model are presented in Table 8. It is evident that in all types of time-frequency images, the fusion of textural features extracted out of these features has demonstrated better results as compared to the individual textural features extraction method, *i.e.*, LTP, LDP, GLTP, and LPQ.

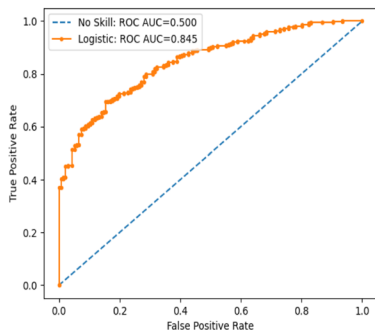
The fusion of the textural features, *i.e.*, LTP, LPQ, GLTP, and LDP has achieved a testing accuracy of more than 90% with the SVM model for every time-frequency image. This accuracy has been achieved on 20% testing data excluding 80% training data. Table 8 illustrates the maximum values observed for sensitivity, specificity, testing accuracy, precision, and F1-score achieved by Gammatonegram against the fusion of extracted textural features. The models with Spectrogram, Scalogram, and Mel-spectrogram images have recorded 0% specificity which indicates that these models are unable



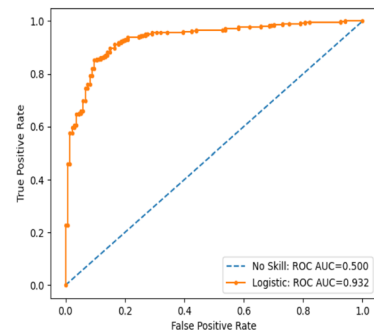
(a) Spectrogram with fusion textural features



(b) Mel-spectrogram with fusion textural features



(c) Scalogram with fusion textural features



(d) Gammatonegram with fusion textural features

Fig. 8 ROC Curve of (a) Spectrogram (b) Mel-spectrogram (c) Scalogram and (d) Gammatonegram with Textural Features Fusion Extraction Method

to identify the true negative cases. The model with Gammatonegram image extraction from PCG signals has demonstrated superior performance than the alternative models consisting of Spectrogram, Mel-spectrogram, and Scalogram. Further, the fusion of all four textural features, *i.e.*, LTP, LDP, GLTP, and LPQ has performed well exhibiting approximately 93% testing accuracy and F1-scores.

In this work, experiments have also been performed with other machine learning and deep learning algorithms over a fusion of all the four structural characteristics of texture extracted from the Gammatonegram image of PCG signals for the purpose of having comparison as demonstrated in Table 9. Out of linear, polynomial, gaussian, and RBF kernels, polynomial kernel performed the best in LSSVM and Fuzzy SVM. The corresponding results are exhibited in Table 9. It can be concluded that CNN performance was competitive, but the proposed model's testing accuracy with the SVM method used for classification was the greatest of all the algorithms.

The confusion matrices of various models are presented in Fig. 9, which explain that the performance of the proposed model is better than all other models. Figure 10 highlights the ROC AUC curves of all the experiments performed which enable to have a comparison with the proposed model. After analysing all the ROC AUC curves, it can be concluded that the best AUC was derived from XGBoost and the proposed model.

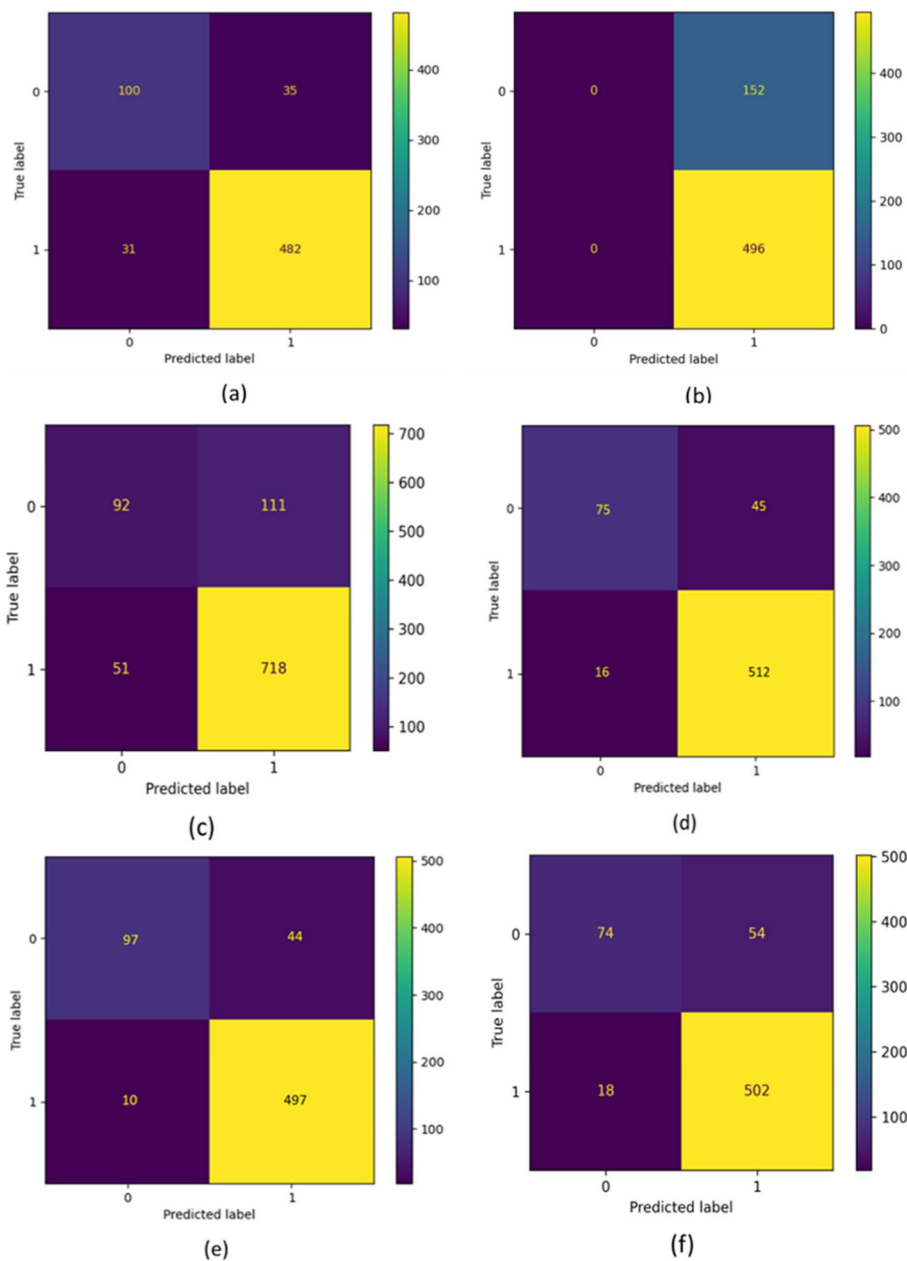


Fig. 9 Confusion Matrix of (a) LSSVM, (b) Fuzzy-SVM, (c) Decision Tree, (d) Random Forest, (e) XGBoost, (f) KNN, (g) ANN, (h) CNN, (i) ResNet50, (j) VGG16, (k) Densenet121, (l) SVM (Proposed model) over Gammatonegram Images with Fusion Textural Features

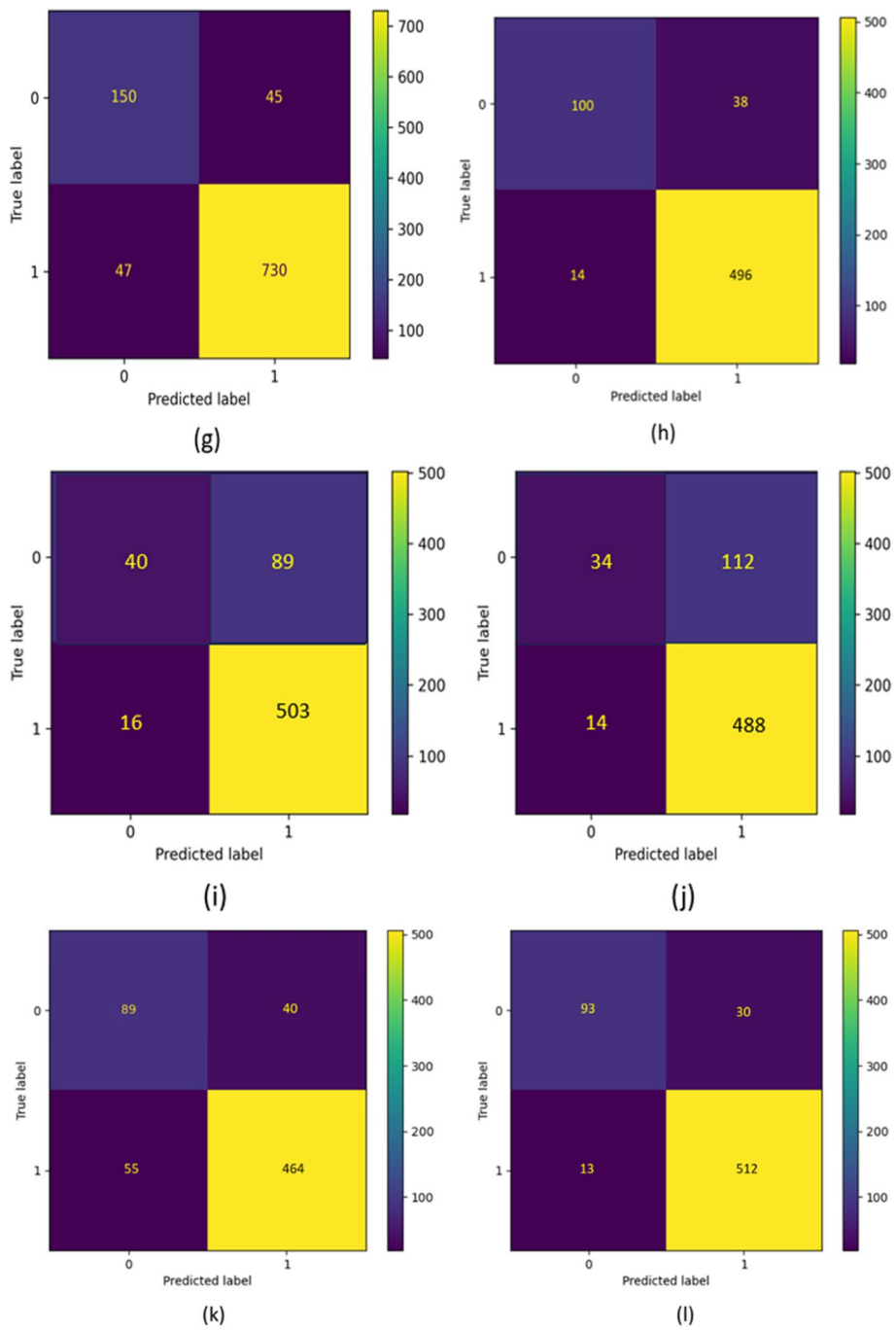


Fig. 9 (continued)

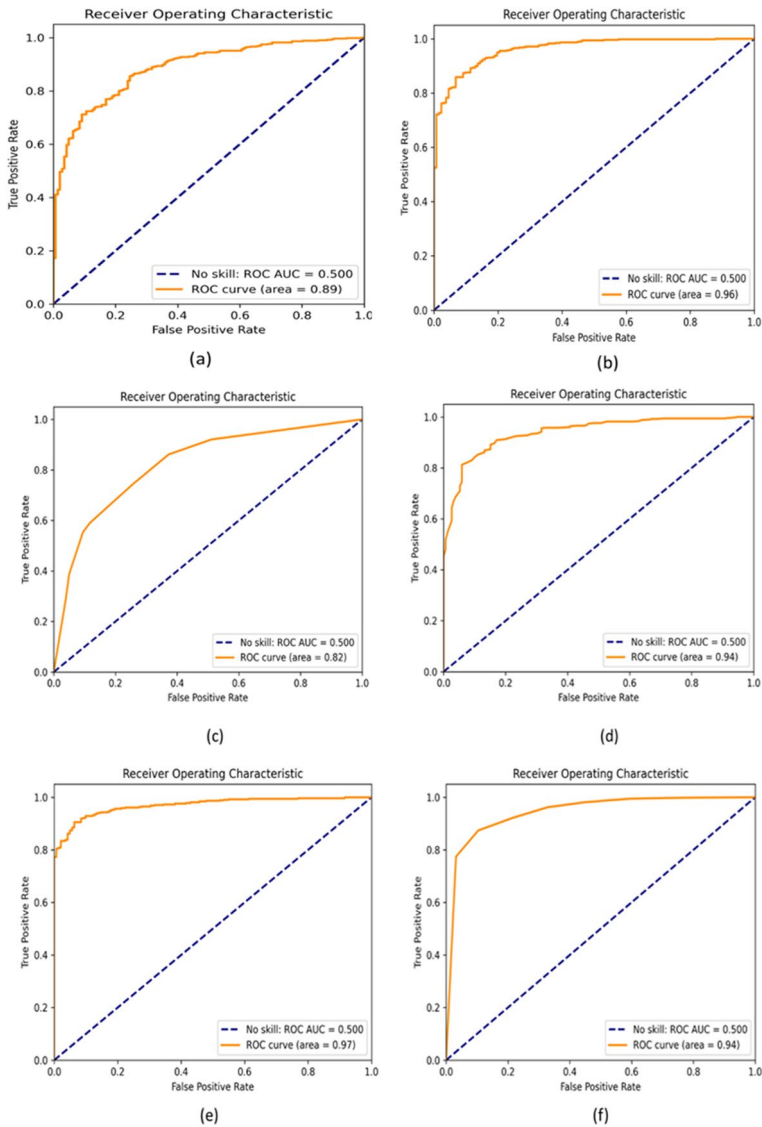


Fig. 10 ROC AUC Curve of (a) LSSVM, (b) Fuzzy-SVM, (c) Decision Tree, (d) Random Forest, (e) XGBoost, (f) KNN, (g) ANN, (h) CNN, (i) ResNet50, (j) VGG16, (k) Densenet121, (l) SVM (Proposed model) over Gammatonegram Images with Fusion Textural Features

5.4 Significance over Peer Approach Results

It can be observed from the literature review presented at a glance in Table 2 that detailed performance results are missing for their classification model. Only one or two performance metrics have been calculated to assess the model's efficiency. The F1-score metric have been found highly valuable in the context of datasets that exhibit imbalanced class distribution. The outcomes of two metrics, namely, precision and

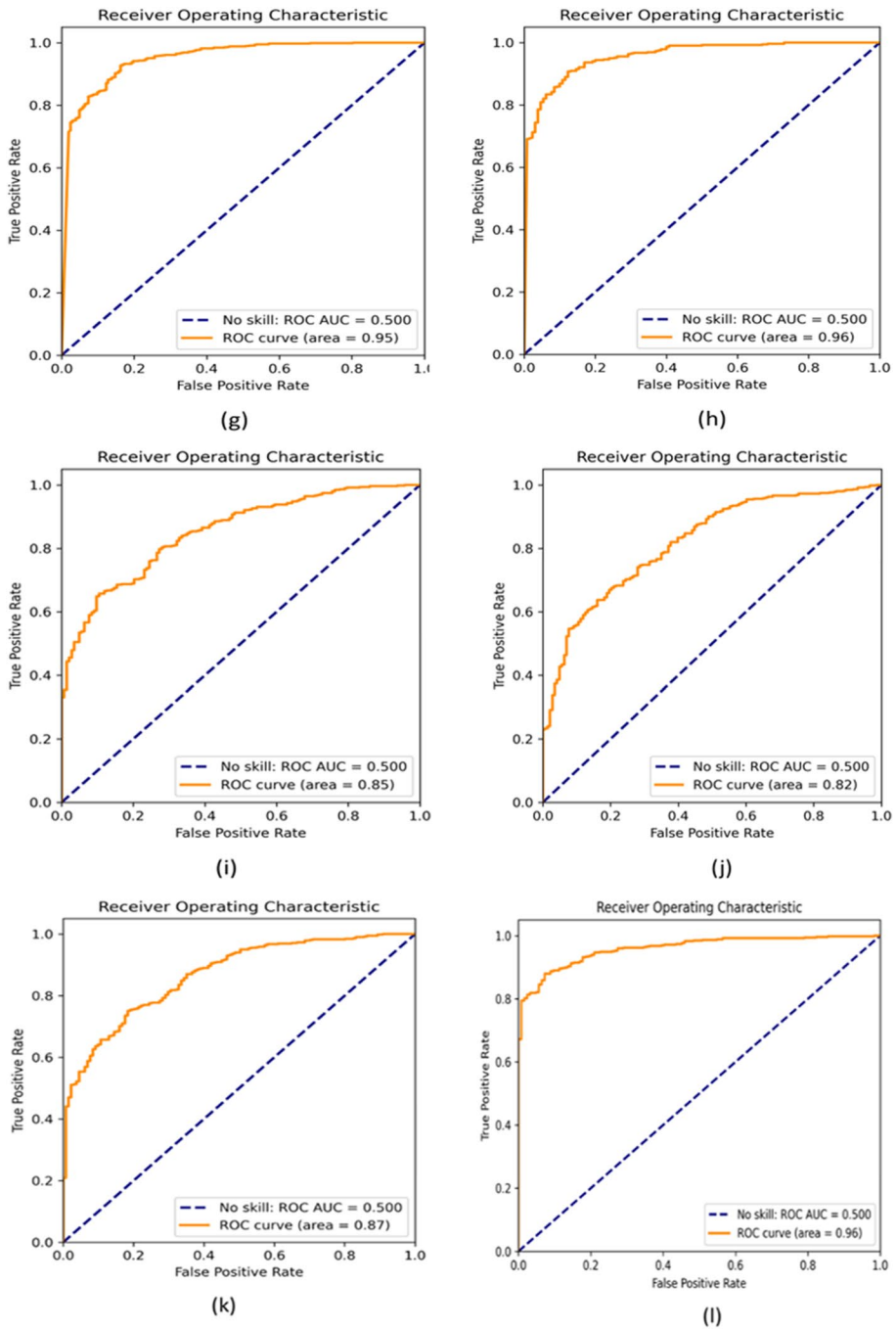


Fig. 10 (continued)

sensitivity (also referred to as recall), are equilibrated with the usage of F1-score. In other words, the metric in question has achieved equilibrium between positive and negative samples, thus, rendering it a superior metric for assessing the efficacy of a machine learning model in scenarios where false negative and false positive values hold significant implications for the real-world issues, as it happens in the detection of any disease problem. None of the peer researchers have calculated the F1-scores of their models as this metric.

The proposed work presents a detailed performance measure over the testing set. The results derived from this study have shown a considerable improvement over the other methods in this area. This can be easily observed from Table 3.

6 Conclusion and future scope

The unsegmented PCG signals were obtained from a reference dataset, PhysioNet CinC 2016. A novel variant to graphically or pictorially convey the information of the PCG signal named Gammatonegram was explored. The textural features of Gammatonegram images were then extracted, and named as LTP, LDP, GLTP, and LPQ. The experimental results showed that extracting the fusion of textural elements from the Gammatonegram representation of PCG signal significantly improved the overall performance as compared to Spectrogram, Mel-spectrogram, and Scalogram time-frequency pictorial portrayal of the dataset. The image textural feature extraction technique used in the proposed work has performed well on the heart sound images as compared to the features extracted by peer researchers in previous studies. Further, Gammatonegram image extraction from heart sounds, which has not been explored, yet in this field, resulted in better performance than other heart sound images. However, certain other image extraction techniques can be used in future work. Further, exploration of various textural characteristics and their fusion variations can potentially yield better results in the classification of heart sounds.

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Data Availability Statement The data which supported the findings of this study is publicly available in The PhysioNet/Computing in Cardiology Challenge 2016 at <https://physionet.org/content/challenge-2016/1.0.0/>.

Declarations

Conflicts of Interest All the authors certify that they have no affiliation with any organization or entity, and any financial or non-financial interest in the subject matter or materials discussed in this manuscript.

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